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salvador

A code for abundance determination through spectral synthesis

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ABSTRACT

BLABLA

Key words. techniques: spectroscopic – methods: data analysis – stars: abundances – stars: atmospheres – stars: rotation

1. Introduction

The chemical composition of stellar photospheres is one of the most powerful diagnostics in stellar and Galactic astrophysics. Chemical abundances trace the nucleosynthetic history of the gas from which stars formed, the mixing processes that modify the surface composition along stellar evolution, and the self-enrichment processes responsible for the multiple populations observed in globular clusters (GCs; e.g., Bastian & Lardo 2018; Gratton et al. 2019). In the last two decades, large spectroscopic surveys such as APOGEE (Majewski et al. 2017), the Gaia-ESO Survey (Gilmore et al. 2022), and GALAH (Buder et al. 2021), together with multi-object and integral-field spectrographs such as FLAMES (Pasquini et al. 2002) and MUSE (Bacon et al. 2010), and very high-resolution spectrographs such as ESPRESSO (Pepe et al. 2021), have provided spectra for hundreds of thousands of stars, with resolving powers ranging from $R \sim 3000$ to more than 100 000. This wealth of data requires analysis tools that are automatic and reproducible, while being flexible enough to deal with spectral features of very different nature: weak and strong atomic lines, blended features, molecular bands, and transitions affected by hyperfine or isotopic splitting.

Chemical abundances are commonly derived following two approaches. In the first one, the equivalent widths (EWs) of isolated lines are measured, either by direct integration or by fitting analytical profiles (e.g., with DAOSPEC or ARES; Stetson & Pancino 2008; Sousa et al. 2007, 2015), and compared with the theoretical curve of growth computed for a given model atmosphere, as done for instance by MOOG (Sneden 1973), GALA (Mucciarelli et al. 2013), and FAMA (Magrini et al. 2013). This approach is fast and naturally suited to the spectroscopic derivation of the atmospheric parameters through the excitation and ionization balance. However, it is restricted to unblended lines with a well-defined profile, and its reliability decreases at low and intermediate spectral resolution, in crowded spectral regions (as in metal-rich or cool stars), for strong lines with extended damping wings, and for features affected by hyperfine or isotopic splitting or by molecular blends. In the second approach, the observed spectrum is directly compared with synthetic spectra. Several codes adopt this strategy, either by computing the synthetic spectra on the fly, as in SME (Valenti & Piskunov 1996; Piskunov & Valenti 2017) and its Python implementation PySME (Wehrhahn et al. 2023), the synthesis driver of MOOG, BACCHUS (Masseron et al. 2016) and TSFitPy (Gerber et al. 2023), both based on Turbospectrum (Alvarez & Plez 1998; Plez 2012), iSpec (Blanco-Cuaresma et al. 2014; Blanco-Cuaresma 2019), and FASMA (Tsantaki et al. 2018), or by interpolating within precomputed grids or libraries, as in MyGIsFOS (Sbordone et al. 2014), SP_Ace (Boeche & Grebel 2016), and the APOGEE pipeline ASPCAP (García Pérez et al. 2016), which relies on the FERRE code (Allende Prieto et al. 2006). Data-driven methods, such as The Cannon (Ness et al. 2015) and The Payne (Ting et al. 2019), represent a further step towards the fast analysis of very large samples. Many of these tools determine atmospheric parameters and chemical abundances simultaneously, often by fitting extended spectral regions.

Despite this variety of tools, chemical abundances derived from the same spectra by different codes and groups can differ significantly, because of differences in the adopted model atmospheres, atomic data, continuum placement, line selection, and treatment of the line broadening (Smiljanic et al. 2014; Hinkel et al. 2016; Jofré et al. 2019). A full control of each step of the analysis, line by line, therefore remains essential, in particular when small abundance variations must be detected, as in the case of the multiple populations of GCs, or when the abundances rely on a few, possibly blended or saturated, spectral features.

In this paper we present salvador[1], a Python code designed to derive chemical abundances through spectral synthesis on a line-by-line basis. For each line, salvador performs a χ^2 minimization between the observed spectrum and a set of synthetic spectra in which only the abundance of the element under analysis is varied. The synthetic spectra are computed on the fly with the SYNTHE code (Kurucz 1993b; Sbordone et al. 2004; Kurucz 2005), adopting one-dimensional, plane-parallel, local thermodynamic equilibrium (LTE) ATLAS9 model atmospheres (Kurucz 1993a, 2005) computed with the opacity distribution functions of the KOALA database (Mucciarelli et al. 2026), both managed through the Python code K2 (Mucciarelli et al., in

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[1] The code is publicly available at https://github.com/DeimerAlvarez/salvador.

prep.). Alternatively, salvador can use a grid of precomputed synthetic spectra, calculated with any spectral synthesis code. salvador does not derive the atmospheric parameters, which must be known beforehand, for instance from photometry or from the spectroscopic analysis described in Sect. 7. The code is designed for the analysis of FGK stars, both in the optical and in the infrared domain, and it can handle spectra with resolving powers from $R \sim 3000$, as those provided by MUSE, up to the very high resolution of ESPRESSO. Previous versions of salvador have already been employed to derive chemical abundances in the GCs ω Centauri and M 54 (Alvarez Garay et al. 2024, 2026).

Besides the standard derivation of abundances from the line profiles, salvador implements a set of features that, to our knowledge, are not combined within a single code:

1. the abundance variation can be mimicked by an equivalent variation of the oscillator strength ($\log gf$) of the transitions of the species under analysis. In this way, the line opacity is modified while the chemical composition adopted in the synthesis, and therefore the ionization and molecular equilibria, the electron pressure, and the continuous opacity, remain those of the model atmosphere. The synthetic spectrum is thus fully consistent with the adopted model atmosphere, also for elements that are important electron donors (e.g., Mg, Si, and Fe) or that take part in the molecular equilibrium (e.g., C, N, and O);
2. the spectral resolution can be estimated automatically for each line, by comparing the full width at half maximum of the observed line with that of the intrinsic (unbroadened) synthetic profile. This is particularly useful for multi-object and integral-field spectrographs, where the resolving power can vary with wavelength and across the detector;
3. the abundances of strong, saturated lines can be derived from their wings only, excluding the line core, which is almost insensitive to the abundance and can be affected by departures from LTE;
4. isotopic ratios, such as ^{12}C/^{13}C and ^{6}Li/^{7}Li, can be measured;
5. the projected rotational velocity ($v \sin i$) can be derived from individual lines, either simultaneously with the abundance or at a fixed abundance;
6. the uncertainties arising from the atmospheric parameters are computed line by line by repeating the full analysis with the perturbed parameters, and they are propagated keeping their sign, so that the sensitivities common to the element under analysis and to iron partially cancel out in the [X/Fe] abundance ratios. The internal uncertainties are estimated through Monte Carlo simulations based on the signal-to-noise ratio measured around each line.

salvador can be run both in an interactive mode, which allows the user to inspect and, when needed, adjust every fit (normalization, wavelength shift, fitting region, and resolution), and in a fully automatic mode, in which several lines can be analyzed in parallel.

Although salvador is designed and tested for FGK stars, it can in principle be applied also to M giants, since the TiO and H_2O line lists, which dominate the opacity in the spectra of the coolest stars, can be included in the computation of the synthetic spectra. However, because of the huge number of molecular transitions involved, the inclusion of these line lists makes the computation of each synthetic spectrum very time consuming, and we therefore recommend this option only for specific cases (Sect. 3.1).

This paper is organized as follows. In Sect. 2 we give an overview of salvador and of its input files and running modes. In Sect. 3 we describe the computation of the synthetic spectra. In Sect. 4 we detail the analysis of the individual lines, and in Sect. 5 the estimate of the uncertainties. In Sect. 6 we describe the implementation and the performance of the code. In Sect. 7 we present the new version of GALA, which derives the atmospheric parameters. In Sect. 8 we validate salvador against synthetic spectra and benchmark stars, and in Sect. 9 we summarize our work.

2. Overview of the code

salvador is designed to measure, for each star, a list of spectral features defined by the user, keeping the analysis of every feature independent of the others. The main steps of the analysis are summarized in Fig. 1. For each observed spectrum, the user provides the atmospheric parameters (effective temperature $T_{\rm eff}$, surface gravity $\log g$, metallicity [M/H], and microturbulent velocity ξ), the chemical mixture of the model atmosphere, the radial velocity (RV) of the star, and a line list specifying, for each feature, the kind of analysis to be performed, the wavelength and the species of the transition, the spectral resolution, the limits of the region where the fit is performed, and the initial abundance. The chemical mixture is chosen among five values of the α-element enhancement, $[\alpha/{\rm Fe}] = -0.4$, -0.2, 0.0, $+0.2$, and $+0.4$ dex, matching the sampling of the KOALA grid of ATLAS9 opacity distribution functions (ODFs; Mucciarelli et al. 2026). Optionally, the uncertainties of $T_{\rm eff}$, $\log g$, and ξ can be given, in which case the corresponding uncertainties on the abundances are computed (Sect. 5.2).

The kind of analysis is set for each feature by a keyword, listed in Table 1. All the keywords share the same core (Sect. 4), that is, the comparison between the observed feature and a set of synthetic spectra that differ only in the quantity to be measured: the abundance of the element (or, equivalently, the $\log gf$ of its transitions), an isotopic ratio, or the projected rotational velocity. A file of abundance variations, associated with each spectrum, allows the user to change the abundance of any element with respect to the chemical mixture of the model atmosphere, or the isotopic composition of a species, in all the synthetic spectra of that star. This is useful, for instance, to include in the synthesis the known abundances of the elements that contribute to the blends of a given feature, as done by Alvarez Garay et al. (2024), who fixed the N abundance when measuring Al from the doublet at 7835–7836 Å, contaminated by CN lines.

salvador can be run in two modes. In the automatic mode, all the features are analyzed without any intervention by the user, and several features can be analyzed in parallel (Sect. 6). In the interactive mode, the fit of each feature is displayed and the user can accept or reject it, or modify the ingredients of the fit (Sect. 4.9). In both modes, the synthetic spectra can be computed on the fly (Sect. 3.1) or taken from a grid of precomputed synthetic spectra (Sect. 3.2).

For each spectrum, salvador provides a file with the results of each feature (abundance, EW, excitation potential χ, $\log gf$, and the uncertainties described in Sect. 5), a document with the plots of all the fits, and diagnostic plots of the Fe abundances as a function of χ, reduced EW, and wavelength, useful to check the adopted $T_{\rm eff}$ and ξ. The final abundance ratios, [Fe/H] and [X/Fe], of all the stars and their uncertainties are collected in a single table. The solar reference abundances can be chosen among those by Caffau et al. (2011) and Lodders (2010), which is the default and the reference mixture of the KOALA models

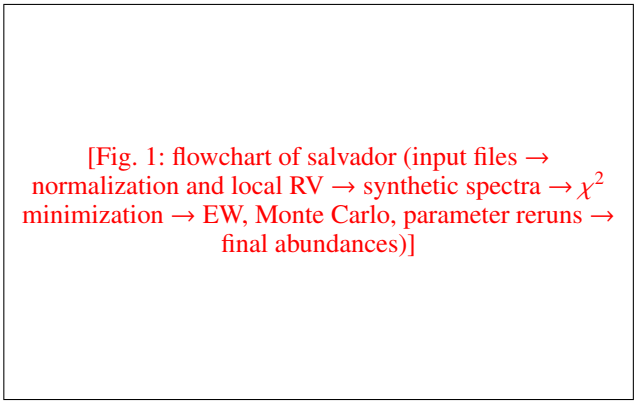


Fig. 1. Schematic workflow of salvador.

Table 1. Keywords defining the kind of analysis of each feature.

Keyword	Analysis
ABU	Abundance from the line profile (Sect. 4.3).
ABUF	As ABU, but the EW includes all the transitions of the species in the feature (e.g., the Li doublet at 6708 Å).
ABU_GF	As ABU, but the abundance variation is mimicked by a variation of $\log gf$ (Sect. 4.4).
ABU2	Abundance from the wings of a strong line (Sect. 4.5).
MOLEC	Abundance from a molecular feature (Sect. 4.6).
ISOR	Isotopic ratio (Sect. 4.7).
ROTAT	$v \sin i$, with the abundance free or fixed (Sect. 4.8).

(Mucciarelli et al. 2026), Grevesse & Sauval (1998), Asplund et al. (2009), and Magg et al. (2022).

3. Synthetic spectra

3.1. Synthetic spectra computed on the fly

In its default configuration, salvador computes the synthetic spectra with the SYNTHE code (Kurucz 1993b; Sbordone et al. 2004; Kurucz 2005), through the Python code K2 (Mucciarelli et al., in prep.), which manages both ATLAS9 and SYNTHE. For each star, K2 computes a one-dimensional, plane-parallel, LTE ATLAS9 model atmosphere (Kurucz 1993a, 2005) with the exact $T_{\rm eff}$, $\log g$, and [M/H] of the star, adopting the ODFs of the KOALA database (Mucciarelli et al. 2026) for the chosen [α/Fe] and interpolating them in [M/H] when needed. The KOALA ODFs are computed with the solar mixture by Caffau et al. (2011) and Lodders (2010) and are available for ξ = 0, 1, 2, 4, and 8 km s^{-1} (Mucciarelli et al. 2026). [confirm with A.M.: which ODF microturbulence is adopted by K2 for a given ξ] Each model atmosphere is computed only once and reused for all the features of the star and for all the following runs.

SYNTHE computes the emergent spectrum at a resolving power of 600 000 in a spectral window centered on the feature, whose width is chosen by the user (a few Å are usually enough) and is automatically enlarged when the fitting region requires it. The atomic and molecular line lists are those of the Kurucz/Castelli database[2] [or other line lists: specify those adopted by default in K2]. The line lists of TiO and H_2O can be included on request. The unbroadened synthetic spectra are convolved by salvador with a Gaussian instrumental profile with the resolving power of the feature (Sect. 4.2), using the PyAstronomy package (Czesla et al. 2019), and resampled on the wavelength grid of the observed spectrum.

The variations of the abundances (or of the isotopic ratios) required during the analysis are passed to K2 together with the variations given by the user in the file of abundance variations (Sect. 2). Each unbroadened synthetic spectrum is stored with a name that encodes all the parameters that define it (atmospheric parameters, chemical mixture, spectral window, and all the abundance variations), so that a synthetic spectrum is computed only once and reused whenever the same spectrum is needed, either in the same run or in subsequent ones.

As discussed in Sect. 1, the TiO and H_2O line lists allow, in principle, the analysis of M giants, down to the lower boundary of the KOALA grid, $T_{\rm eff}$ = 3750 K (Mucciarelli et al. 2026), at the cost of a dramatic increase of the computing time of each synthetic spectrum.

3.2. Grids of precomputed synthetic spectra

Alternatively, salvador can work with a grid of precomputed, continuum-normalized, unbroadened synthetic spectra, provided by the user and calculated with any spectral synthesis code. For each feature, salvador selects the synthetic spectra with exactly the atmospheric parameters, [M/H], and [α/Fe] of the star, computed with different abundances of the element under analysis, and covering the wavelength of the feature. In this case, no new synthetic spectrum can be computed during the analysis. Therefore, the abundances that can be measured are limited to the range covered by the grid, and the features whose χ^2 minimum falls at the edge of the grid are rejected in the automatic mode. The EWs are measured directly on the best-fit synthetic spectrum, while χ and $\log gf$ of the transitions are not available. The uncertainties due to the atmospheric parameters (Sect. 5.2) can be computed only if the grid also includes the synthetic spectra with the perturbed parameters.

4. Analysis of the spectral features

4.1. Continuum normalization and local wavelength alignment

The observed spectrum, read from a FITS or ASCII file, is cleaned of the pixels with non-finite values and shifted to the rest frame with the RV provided by the user. Three options are available for the continuum normalization. The spectrum can be used as it is, if already normalized; it can be rigidly rescaled to a synthetic spectrum computed with the parameters of the star, by dividing it by the median observed flux at the wavelengths where the synthetic spectrum is within 0.5% of its maximum (i.e., at the continuum); or it can be normalized by using a set of continuum windows provided by the user. In the latter case, the pixels of each window that deviate by more than 5σ from the median (with σ estimated from the median absolute deviation) are removed, splitting the window into clean sub-windows, the mean flux in each sub-window defines a continuum point, and the continuum is obtained by interpolating these points with a cubic spline (or linearly when fewer than four points are avail-

[2] https://wwwuser.oats.inaf.it/castelli/linelists.html

able). Since in cool or metal-rich stars the continuum windows are rarely free of weak absorption features, the normalized spectrum is then multiplied by the mean level of the synthetic spectrum of the star in the same windows, interpolated in the same way. This procedure takes into account the pseudo-continuum predicted by the synthesis and avoids an underestimate of the continuum level in crowded spectral regions.

Residual wavelength offsets between the observed and the synthetic spectra, due for instance to uncertainties in the RV or in the wavelength calibration, can be corrected locally. The center of the observed feature is measured as the mean between the center of a Gaussian fit and the midpoint of the two wavelengths at which the profile crosses its half depth. If the offset between this center and the laboratory wavelength is smaller than a tolerance set by the user (in pixels), the observed spectrum is shifted accordingly for the analysis of that feature.

4.2. Spectral resolution

The resolving power adopted to broaden the synthetic spectra can be specified for each feature in the line list. This allows the user to account for the variation of the resolving power with wavelength, or among different setups of the same instrument. Moreover, salvador can estimate the resolving power directly from each observed feature. The full width at half maximum of the observed profile, $\mathrm{FWHM_{obs}}$, is measured as the mean of the values obtained from a Gaussian fit and from the half-depth crossings of a spline interpolation of the profile. The same measurement is performed on the unbroadened synthetic spectrum, after isolating the profile of the main transition from the adjacent ones, providing the intrinsic width $\mathrm{FWHM_{syn}}$, which includes the thermal, microturbulent, and damping broadening. Assuming that the instrumental profile is Gaussian, and approximating the intrinsic profile as Gaussian as well, the instrumental width and the corresponding resolving power are

$$\mathrm{FWHM_{ins}} = \sqrt{\mathrm{FWHM_{obs}^2} - \mathrm{FWHM_{syn}^2}}\,, \qquad R = \frac{\lambda}{\mathrm{FWHM_{ins}}}\,. \tag{1}$$

The estimated R replaces the input value only if the relative difference between the two is larger than 2% and smaller than a threshold chosen by the user, in order to protect the analysis against spurious estimates (e.g., in weak, noisy, or blended features). Since no macroturbulent broadening is included in the synthesis, the resolving power estimated in this way also absorbs the macroturbulence of the star, providing an effective Gaussian broadening kernel for the feature.

4.3. Abundance determination

For a given feature, the abundance of the element is derived by minimizing

$$\chi^2(A) = \sum_i \frac{\left[F_{\mathrm{obs},i} - F_{\mathrm{syn},i}(A)\right]^2}{F_{\mathrm{syn},i}(A)}\,, \tag{2}$$

where $F_{\mathrm{obs},i}$ and $F_{\mathrm{syn},i}(A)$ are the observed and synthetic normalized fluxes at the i-th pixel of the fitting region, and A is the abundance of the element, $A(\mathrm{X}) = \log(N_\mathrm{X}/N_\mathrm{H}) + 12$. For a spectrum dominated by photon noise, the variance of the normalized flux at each pixel is $\sigma_i^2 \simeq F_i/(\mathrm{S/N})_\mathrm{c}^2$, where $(\mathrm{S/N})_\mathrm{c}$ is the signal-to-noise ratio in the continuum. Therefore, Eq. (2) corresponds, apart from the constant factor $(\mathrm{S/N})_\mathrm{c}^2$, which does not affect the position of the minimum, to the χ^2 of a photon-noise-limited spectrum, with the variance of each pixel estimated from the model (Pearson's χ^2). Additional noise sources, such as readout noise and sky subtraction, are not taken into account. The limits of the fitting region are given by the user or determined automatically from the observed profile.

The synthetic spectra needed to bracket the χ^2 minimum are computed adaptively, to keep their number as small as possible (Fig. 1). The first synthetic spectrum is computed with the initial abundance given in the line list. The integrated line depression, $\int(1-F)\,\mathrm{d}\lambda$, of the observed feature is compared with that of the synthetic one, to establish whether the abundance must be increased or decreased. A second synthetic spectrum is then computed with an abundance variation of 0.4 dex in that direction. If the observed line depression lies between the two synthetic ones, a third synthetic spectrum is computed at the intermediate abundance; otherwise, two synthetic spectra are computed at 0.6 and 0.8 dex from the initial abundance. In both cases, three synthetic spectra separated by 0.2 dex are available. The χ^2 of each of them is computed and, if the minimum is not bracketed, the set of three synthetic spectra is moved in steps of 0.2 dex towards the decreasing χ^2, until the minimum is bracketed or the abundance differs from the initial one by more than a threshold chosen by the user (3 dex by default), in which case the feature is rejected.

Once the minimum is bracketed, a fine grid of synthetic spectra with a step of 0.01 dex is built by linear interpolation in abundance of the synthetic fluxes, pixel by pixel, between the synthetic spectra adjacent to the one with the lowest χ^2, and the abundance that minimizes Eq. (2) on this grid is adopted. The spacing of 0.2 dex between the computed synthetic spectra is a compromise between the number of syntheses and the accuracy of the linear interpolation of the flux, which degrades for larger spacings, in particular for saturated lines. Finally, the synthetic spectrum with the best-fit abundance is computed directly with SYNTHE, together with two synthetic spectra at ±0.2 dex, which are shown in the plot of the fit (Fig. 2).

After the abundance is determined, the EW of the feature is computed by K2 from a synthetic spectrum that includes only the transition under analysis (and its hyperfine or isotopic components) with the best-fit abundance. Alternatively, all the transitions of the same species falling in the feature can be included, as required for instance by the Li resonance doublet. The EW is not used to derive the abundance, but it provides the reduced EW, $\log(\mathrm{EW}/\lambda)$, which is used, together with χ, to select the features entering the final abundance (Sect. 4.10) and to build the diagnostic plots of Fe.

4.4. Abundances from the variation of the oscillator strength

When the abundance of an element is varied only in the spectral synthesis, the thermal and pressure structure of the model atmosphere, the electron pressure, and the continuous opacity remain those computed for the original chemical composition. This approach does not introduce significant errors as long as the abundance variation has a negligible impact on the opacity and on the ionization structure of the atmosphere (Mucciarelli et al. 2026). However, the spectral synthesis code recomputes the ionization and molecular equilibria with the new abundance, so that the synthetic spectrum is not computed with the same chemical composition as the model atmosphere. This inconsistency can become relevant for large abundance variations of the elements

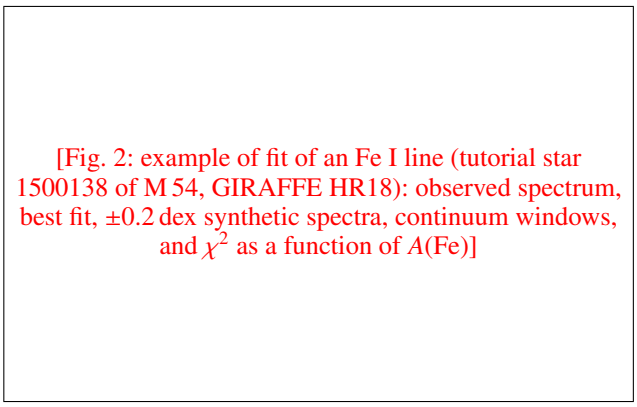


Fig. 2. Example of the fit of a Fe I line.

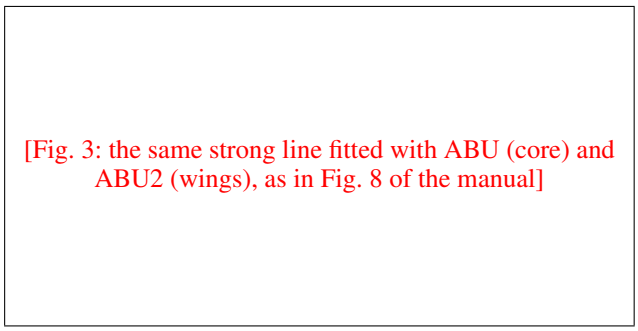


Fig. 3. Comparison between the fit of a strong line performed on the core (keyword ABU, left) and on the wings (keyword ABU2, right).

that provide most of the free electrons, such as Mg, which is the main electron donor in the atmospheres of FGK stars (Mucciarelli et al. 2026), or of the elements involved in the molecular equilibrium, such as C, N, and O.

salvador offers the possibility to avoid this inconsistency. In LTE, the line absorption coefficient of a transition is proportional to the product $gf\,N_X$ of its oscillator strength and of the number density of the absorbing element (e.g., Gray 2005). Therefore, as far as the line opacity is concerned, a variation Δ of $\log gf$ of the transitions of the species under analysis is equivalent to a variation Δ of A(X). With the keyword ABU_GF, the synthetic spectra are computed by varying the $\log gf$ of these transitions instead of the abundance, while the chemical composition adopted in the synthesis is the same as that of the model atmosphere. The abundance is then $A(\mathrm{X}) = A_{\mathrm{model}}(\mathrm{X}) + \Delta$, where $A_{\mathrm{model}}(\mathrm{X})$ is the abundance of the element in the model atmosphere and Δ is the best-fit variation of $\log gf$. In this way, the synthetic spectrum is fully consistent with the adopted model atmosphere. We note that neither approach is equivalent to the computation of a new model atmosphere with the derived abundance, which would require the opacity sampling technique of ATLAS12 (Kurucz 2005) or new ODFs; when the derived abundance differs significantly from that of the model atmosphere, the analysis should be iterated with a more appropriate chemical mixture.

4.5. Strong lines: abundances from the line wings

The core of strong, saturated lines is almost insensitive to the abundance and forms in the outermost layers of the atmosphere, where the assumptions of LTE and of a photospheric temperature structure are less reliable, and where the ATLAS9 models may not reach full convergence (Mucciarelli et al. 2026). On the other hand, the wings of these lines, dominated by collisional damping, are formed in deeper layers and remain sensitive to the abundance. With the keyword ABU2, the χ^2 of Eq. (2) is computed over two separate regions covering the blue and red wings of the line, excluding the core (Fig. 3). The rest of the analysis is the same as described in Sect. 4.3. Since the damping wings depend on the gas pressure, the abundances derived from them are sensitive to $\log g$ and to the chemical mixture of the model atmosphere (Mucciarelli et al. 2026), and an accurate estimate of the uncertainty due to $\log g$ is particularly relevant for these features (Sect. 5.2).

4.6. Molecular features

With the keyword MOLEC, the abundance is derived from a molecular feature (e.g., CH, C_2, or CN bands) by varying the abundance of the element under analysis, as for the atomic lines. The abundances of the other elements forming the molecule can be fixed through the file of abundance variations, for instance the C abundance when N is derived from CN features. For these features, the EW, χ, and $\log gf$ are not computed.

4.7. Isotopic ratios

With the keyword ISOR, salvador measures the ratio between the abundances of two isotopes of the same element, such as ^{12}C/^{13}C from CN features or ^{6}Li/^{7}Li from the Li resonance doublet. The procedure is the same as for the abundances, but the synthetic spectra are computed with different isotopic ratios, with steps of 10 between the computed synthetic spectra (20 for the first step), and the fine grid has a step of 1. The ratio is not allowed to go below unity. The elemental abundance adopted in the synthesis is fixed, and should be derived beforehand, for instance from features insensitive to the isotopic ratio. The isotopic fractions of elements with more than two isotopes can be set in the file of abundance variations.

4.8. Projected rotational velocity

With the keyword ROTAT, salvador derives the projected rotational velocity, $v \sin i$, from a feature, either simultaneously with the abundance of the element or at a fixed abundance. A grid of synthetic spectra is computed with SYNTHE for abundances in steps of 0.1 dex within a range set by the user around the initial abundance (a single abundance if it is kept fixed), and convolved with the instrumental profile. Each of them is then broadened with the rotational profile of Gray (2005), with a linear limb-darkening coefficient of 0.6, as implemented in PyAstronomy (Czesla et al. 2019), for $v \sin i$ values from a minimum to a maximum value (by default 0 and 60 km s^{-1}) in steps of 10 km s^{-1}. Since the rotational broadening is applied to the synthetic spectra already computed, a finer sampling in $v \sin i$ does not require any additional synthesis. The χ^2 of Eq. (2) is computed over this grid, and around its minimum a fine grid with steps of 1 km s^{-1} (within ±20 km s^{-1}) and 0.01 dex (within ±0.2 dex) is built by cubic interpolation (linear when fewer than four nodes are available along an axis). The minimum of χ^2 on the fine grid provides $v \sin i$ and the abundance. Their uncertainties are estimated with a Monte Carlo simulation, by adding to the best-fit spectrum Gaussian noise with a dispersion equal to that of the residuals of the fit, and repeating the search for the minimum (100 realiza-

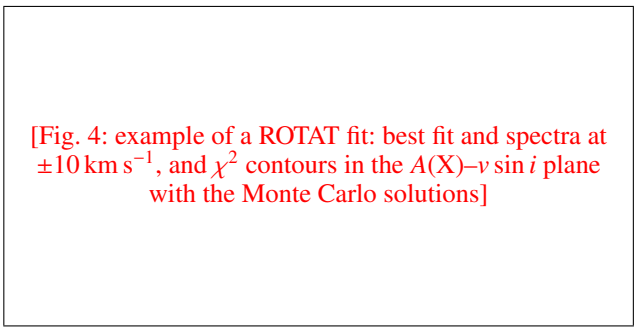


Fig. 4. Example of the simultaneous determination of $v \sin i$ and abundance.

tions by default). As no macroturbulent broadening is included in the synthesis, the derived $v \sin i$ includes any additional broadening not accounted for by the instrumental profile; values comparable to the expected macroturbulent velocity should therefore be regarded as upper limits.

4.9. Interactive mode

In the interactive mode, the fit of each feature is displayed and the user can accept it, reject it, or modify its ingredients: rescale the normalized spectrum by a constant factor, shift it in wavelength, change the limits of the fitting regions, change the resolving power, add or remove continuum windows (the spectrum is then normalized again), or move the set of synthetic spectra by one step towards the decreasing χ^2. After each change the fit is repeated, and the initial configuration can be restored at any time. This mode is especially useful for the analysis of a few stars or of critical features (e.g., weak or blended lines, or lines in crowded spectral regions), and to define the fitting regions to be adopted in the automatic analysis of large samples.

4.10. Final abundances

For each star and species, the abundances of the individual features are combined into a mean value. The features can be selected according to their χ and reduced EW, within limits chosen by the user, to exclude, for instance, strong lines that are more sensitive to ξ and to the damping constants, or features too weak to be reliably measured. A σ-clipping is then applied, adopting the median and the median absolute deviation as estimators of the center and of the dispersion, when at least three features are available. [Fe/H] is derived from the Fe I lines, and the abundance ratios of the other species are provided as [X/Fe]. The absolute abundances A(X) and the [X/H] ratios are also provided.

Different spectra of the same star, such as those obtained with different setups of a multi-object spectrograph, can be analyzed separately and their results merged afterwards, recomputing the final abundances and their uncertainties with the same procedure.

5. Uncertainties

5.1. Internal uncertainties

For the species measured from fewer than ten features, the uncertainty of each feature due to the photon noise is estimated through a Monte Carlo simulation. The signal-to-noise ratio (S/N) is measured within ±20 Å from the feature, either in the continuum windows or, when these are not available, on the flux after a 3σ-clipping that removes the absorption features. Gaussian noise with a dispersion equal to $(\mathrm{S/N})^{-1}$ is added to the best-fit synthetic spectrum and the abundance is derived again from the fine grid of Sect. 4.3. The procedure is repeated 200 times and, after discarding the 5% of the highest and lowest values, the standard deviation of the derived abundances is adopted as the uncertainty. The random numbers are generated from a seed computed from the name of the spectrum, the wavelength, and the species of the feature, so that the results are fully reproducible, independently of the order in which the features are analyzed.

The internal uncertainty of the mean abundance of a species is computed as $\sigma/\sqrt{N}$, where σ is the standard deviation of the abundances of the N features that survived the selection. When only one feature is available, the Monte Carlo uncertainty is adopted.

5.2. Uncertainties due to the atmospheric parameters

When the uncertainties of the atmospheric parameters are provided, the analysis of each feature is repeated by varying, one at a time, $T_{\rm eff}$, $\log g$, and ξ by their uncertainties, while keeping the other parameters fixed. New model atmospheres are computed by K2 for the perturbed parameters. For each feature and parameter p, the difference $\delta^p = A(\mathrm{X})_{p+\Delta p} - A(\mathrm{X})$ is stored with its sign. The variation of $\log g$ is performed for all the features. To save computing time, the variation of ξ is performed only for the features with EW larger than 50 mÅ, which are the most sensitive to this parameter, and for those whose EW is not computed (molecular features, line wings, and isotopic ratios).

For each species, the mean values of $\delta^{T_{\rm eff}}$, $\delta^{\log g}$, and δ^{ξ} over the selected features are computed. The total uncertainties of [Fe/H] and [X/Fe] are then

$$\sigma_{\rm [Fe/H]} = \sqrt{\frac{\sigma^2_{\rm Fe}}{N_{\rm Fe}} + (\delta^{T_{\rm eff}}_{\rm Fe})^2 + (\delta^{\log g}_{\rm Fe})^2 + (\delta^{\xi}_{\rm Fe})^2}\,, \tag{3}$$

$$\sigma_{\rm [X/Fe]} = \left[\frac{\sigma^2_{\rm X}}{N_{\rm X}} + \frac{\sigma^2_{\rm Fe}}{N_{\rm Fe}} + (\delta^{T_{\rm eff}}_{\rm X} - \delta^{T_{\rm eff}}_{\rm Fe})^2 + (\delta^{\log g}_{\rm X} - \delta^{\log g}_{\rm Fe})^2 + (\delta^{\xi}_{\rm X} - \delta^{\xi}_{\rm Fe})^2\right]^{1/2}, \tag{4}$$

where the internal terms σ^2/N are replaced by the squared Monte Carlo uncertainty when only one feature is available. Since the differences are taken with their sign, the uncertainties due to the atmospheric parameters partially cancel out in [X/Fe] for the species that respond to the parameters as Fe does, while they add up for those that respond in the opposite way, as some ionized species with respect to $T_{\rm eff}$. The uncertainties of [X/H] are computed in the same way, without the terms relative to Fe. This approach neglects the covariances among the atmospheric parameters and should therefore be regarded as an upper limit to the real uncertainty.

6. Implementation and performance

salvador is written in Python (≥ 3.10) and relies on NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas (McKinney 2010), Matplotlib (Hunter 2007), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), specutils (Earl et al. 2026), and PyAstronomy (Czesla et al. 2019). It is distributed as a Python

package under the MIT license, and it can be installed with `pip` from the GitHub repository, together with a tutorial and a user manual. The code is organized in modules according to their task (input and output, normalization, synthesis, fitting, rotation, uncertainties, and final abundances), and all the settings of a run are read from a single input file, checked at the start of the run.

Most of the computing time is spent in the calculation of the synthetic spectra. To reduce it, salvador (i) runs K2 within its own Python process; (ii) passes to SYNTHE, for each feature, only the portion of the line lists within 50 Å from the spectral window, which provides synthetic spectra identical to those computed with the complete line lists; (iii) reuses the model atmospheres and the synthetic spectra already computed (Sect. 3.1); and (iv) in the automatic mode, analyzes several features in parallel, each in a separate process with its own working directory. The model atmospheres needed by all the features are computed before the parallel analysis starts, and the results are written in the same order as in the line list, so that the output is identical to that of a sequential run. As an example, the analysis of 25 lines in the GIRAFFE HR18 spectrum of a giant star of M 54 ([Fe/H] = −1.54 dex), with the continuum normalized through continuum windows and the synthetic spectra computed in windows of 5 Å, required 3.4 min with a single process and 52 s with seven parallel processes (a speed-up of a factor of 3.9), on a laptop with an Intel Core Ultra 7 258V processor and 32 GB of RAM. [confirm: synth_out initially empty (all models and synthetic spectra computed from scratch)? errors due to the atmospheric parameters computed?] The speed-up is lower than the number of processes because the total time is set by the slowest features and because the model atmospheres are computed before the parallel analysis starts.

The reproducibility of the results is ensured by the seeded Monte Carlo simulations (Sect. 5.1) and by a regression tool that compares the results of a run, value by value, with those of a reference run. A second tool verifies that the faster calls to K2 provide synthetic spectra and EWs identical to those of the standard calls. A suite of automatic tests, which replaces K2 with the synthetic spectra of reference runs, is executed at every change of the code.

7. A new version of GALA

[to be written with A. Mucciarelli]

8. Validation

[synthetic spectra and benchmark stars]

9. Conclusions

In this paper we presented salvador, a Python code for the determination of chemical abundances through spectral synthesis, designed for the line-by-line analysis of FGK stars in the optical and infrared domains, from the low resolution of integral-field spectrographs such as MUSE ($R \sim 3000$) to the very high resolution of ESPRESSO. For each feature, salvador minimizes the χ^2 between the observed spectrum and a set of synthetic spectra that differ only in the quantity to be measured. The synthetic spectra are computed on the fly with ATLAS9 and SYNTHE through K2, adopting the KOALA opacity distribution functions (Mucciarelli et al. 2026), or taken from a grid of precomputed spectra calculated with any spectral synthesis code. An adaptive search keeps the number of syntheses small, and the minimum is refined on a grid with a step of 0.01 dex obtained by interpolation of the synthetic fluxes.

Besides the standard abundance analysis, salvador combines within a single framework several features that are rarely available together:

- the abundance variation can be mimicked by a variation of the $\log gf$ of the transitions, keeping the chemical composition of the synthesis consistent with that of the model atmosphere;
- the resolving power can be estimated from each observed feature, accounting for the intrinsic width of the line;
- the abundances of saturated lines can be derived from their damping wings only, excluding the line core;
- isotopic ratios and the projected rotational velocity (with the abundance either free or fixed) can be measured;
- the uncertainties due to the photon noise are estimated through reproducible Monte Carlo simulations, and those due to the atmospheric parameters are computed line by line and propagated with their sign into the abundance ratios.

The interactive mode allows full control of every fit, while the automatic mode, with the analysis of several features in parallel, makes the code suitable for large spectroscopic samples.

[Summary of the validation on synthetic spectra and benchmark stars (Sect. 8): accuracy and precision of the abundances, of $v \sin i$, and of the isotopic ratios.]

salvador does not derive the atmospheric parameters, which must be provided by the user. The new version of GALA presented in Sect. 7 fills this gap, and the two codes together provide a complete chain for the chemical analysis of stellar spectra, from the atmospheric parameters to the final abundances and their uncertainties. Future developments of salvador will include the convolution of the synthetic spectra with non-Gaussian instrumental profiles and a continuum normalization based on a polynomial fit of the ratio between the observed and the synthetic spectra. salvador is publicly available on GitHub, together with a user manual and a tutorial.

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